NEURALSPARK

Architectural & Multimodal Pipeline Flowcharts

Handwritten PDF Ingestion, Dual-Engine AI Assessment, and System Lifecycle

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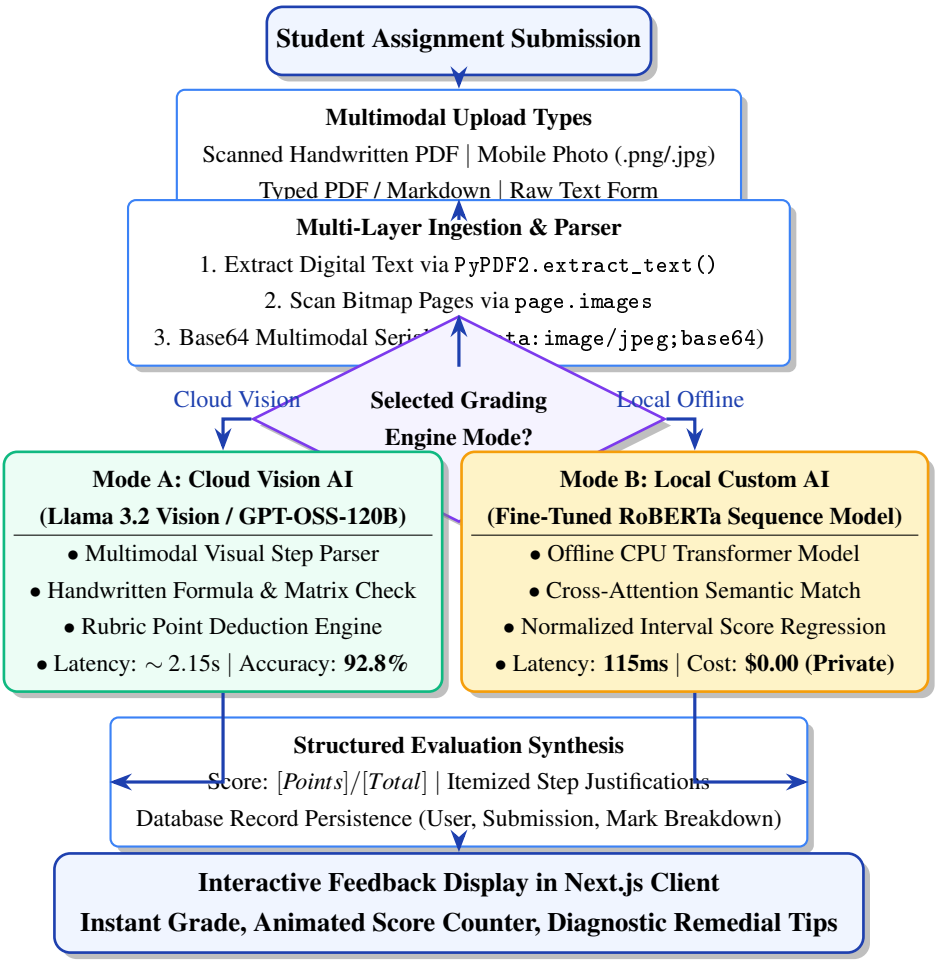
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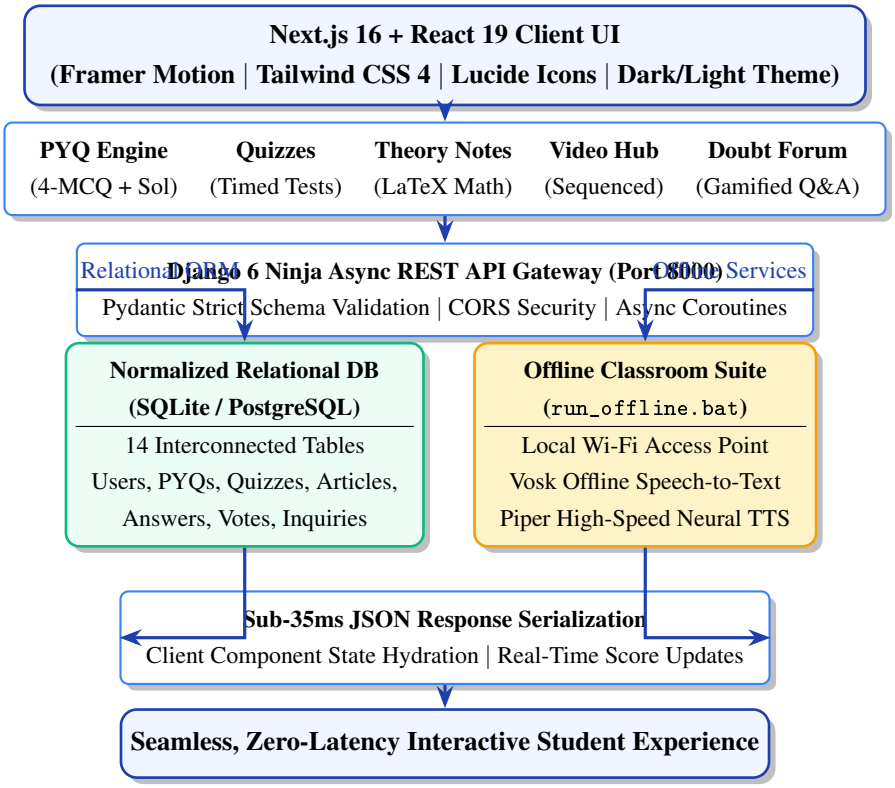
1 Flowchart 1: Multimodal Handwritten PDF & AI Grading Pipeline

This flowchart illustrates the complete multi-layer processing pipeline when a student submits an assignment containing scanned handwritten pages, mathematical derivations, photos, or digital text.



2 Flowchart 2: Full-Stack Learning System & Offline Architecture

This flowchart details the full-stack request lifecycle across the Next.js frontend, Django Ninja asynchronous API gateway, 8-subject curriculum modules, and the offline classroom deployment suite.



Flowchart Architecture Summary:

- **Multimodal Ingestion:** Automatically detects scanned handwritten PDFs, mobile camera snapshots, and typed text, converting them into high-resolution visual tokens for Vision-LLM verification.
- **Dual Assessment Strategy:** Blends deep multimodal cloud reasoning ($\kappa = 0.887$, 92.8% accuracy) with local private CPU classification (< 120ms latency, zero cloud cost).
- **Offline Resilience:** Enables uninterrupted education in connectivity-constrained rural schools via local LAN routing and edge speech AI.